Lab 2 - Greedy regret heuristics

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Description of the problem

Each instance contains nodes with coordinates (x, y) and a cost. Select exactly half of the nodes (rounded up) and connect them into a closed cycle. Minimise **cycle length + sum of the costs of the selected nodes**. Distances are Euclidean, rounded to the nearest integer.

Algorithms

TODO: describe the implemented algorithms and give their pseudocode.

Results

Objective = cycle length + sum of node costs, over 200 runs per method. The best value in each column is in bold.
All best solutions were checked with the solution checker (the same computation as the lecturer's Solution checker.xlsx): **yes**.
Results are identical in all implemented languages: **yes**.

Instance TSPA

Method	Min	Max	Avg	Best solution checked
Lab 1: Random solution	237131	287973	263514	yes
Lab 1: Nearest neighbor (end only)	84362	90075	87477	yes
Lab 1: Nearest neighbor (any position)	72328	77022	74347	yes
Lab 1: Greedy cycle	72590	**75281**	73797	yes
Lab 2: Nearest neighbor (any position), 2-regret	111527	133181	122861	yes
Lab 2: Nearest neighbor (any position), 2-regret + best change	72419	78504	75561	yes
Lab 2: Greedy cycle, 2-regret	112654	130924	121007	yes
Lab 2: Greedy cycle, 2-regret + best change	**72011**	75532	**73350**	yes

Instance TSPB

Method	Min	Max	Avg	Best solution checked
Lab 1: Random solution	188214	231879	206802	yes
Lab 1: Nearest neighbor (end only)	56007	63166	59397	yes
Lab 1: Nearest neighbor (any position)	52915	66164	59475	yes
Lab 1: Greedy cycle	53016	63736	59709	yes
Lab 2: Nearest neighbor (any position), 2-regret	69032	80767	74763	yes
Lab 2: Nearest neighbor (any position), 2-regret + best change	**50485**	**61406**	**55698**	yes
Lab 2: Greedy cycle, 2-regret	69455	78142	73403	yes
Lab 2: Greedy cycle, 2-regret + best change	52906	63326	57751	yes

Running time

Average time of one run of the algorithm only, over the same runs as above. The fastest language in each row is in bold.

Instance TSPA

Method	Python	C	C++	Rust
Lab 1: Random solution	77.6 µs	380 ns	321 ns	**295 ns**
Lab 1: Nearest neighbor (end only)	374.0 µs	**14.0 µs**	20.4 µs	23.5 µs
Lab 1: Nearest neighbor (any position)	24.39 ms	534.2 µs	**481.1 µs**	567.2 µs
Lab 1: Greedy cycle	22.58 ms	687.4 µs	**462.6 µs**	648.7 µs
Lab 2: Nearest neighbor (any position), 2-regret	42.75 ms	**688.7 µs**	2.96 ms	3.21 ms
Lab 2: Nearest neighbor (any position), 2-regret + best change	44.34 ms	**678.8 µs**	2.86 ms	3.14 ms
Lab 2: Greedy cycle, 2-regret	43.19 ms	**942.5 µs**	2.80 ms	1.41 ms
Lab 2: Greedy cycle, 2-regret + best change	43.23 ms	**943.2 µs**	2.78 ms	1.39 ms

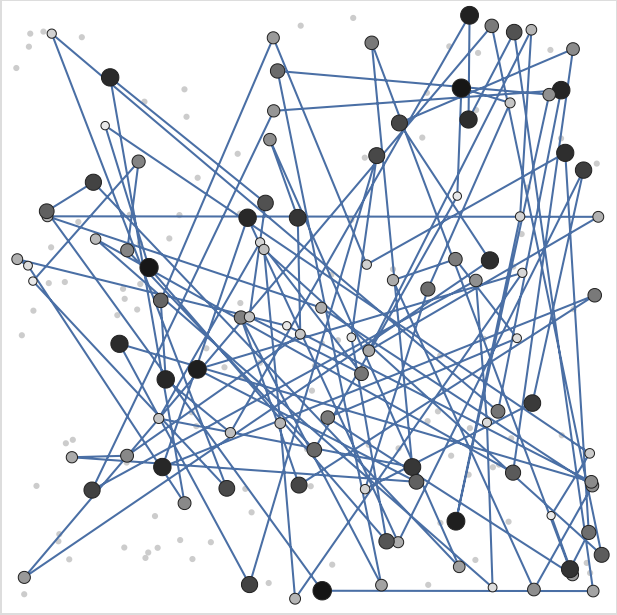
Instance TSPB

Method	Python	C	C++	Rust
Lab 1: Random solution	70.2 µs	**425 ns**	481 ns	500 ns
Lab 1: Nearest neighbor (end only)	360.0 µs	**15.6 µs**	24.0 µs	18.3 µs
Lab 1: Nearest neighbor (any position)	23.51 ms	529.4 µs	**515.9 µs**	561.6 µs
Lab 1: Greedy cycle	21.92 ms	541.8 µs	**464.2 µs**	639.2 µs
Lab 2: Nearest neighbor (any position), 2-regret	44.36 ms	**679.2 µs**	2.78 ms	3.66 ms
Lab 2: Nearest neighbor (any position), 2-regret + best change	43.22 ms	**704.9 µs**	2.91 ms	3.14 ms
Lab 2: Greedy cycle, 2-regret	43.15 ms	**941.7 µs**	2.80 ms	1.48 ms
Lab 2: Greedy cycle, 2-regret + best change	42.78 ms	**941.3 µs**	2.76 ms	1.40 ms

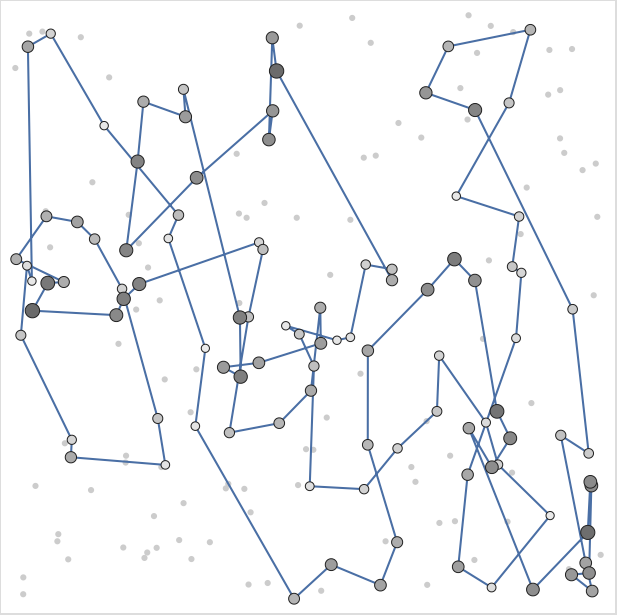
Visualisation of the best solutions

Node colour and size show the node cost (dark and large = expensive). Grey dots are nodes not selected.

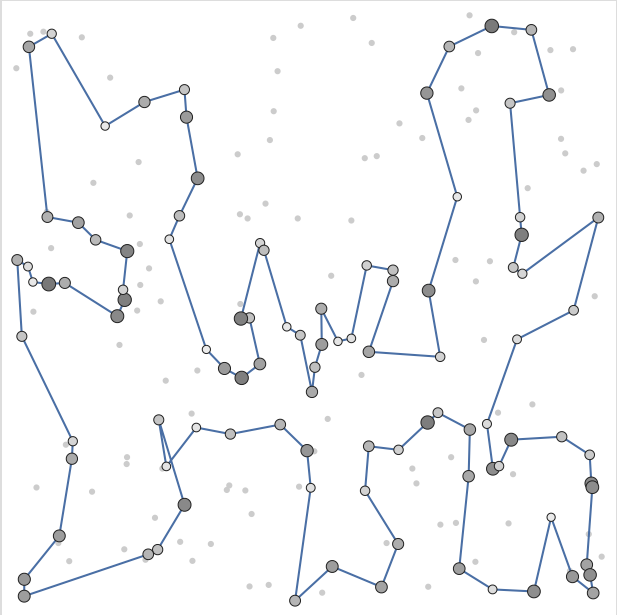
Instance TSPA



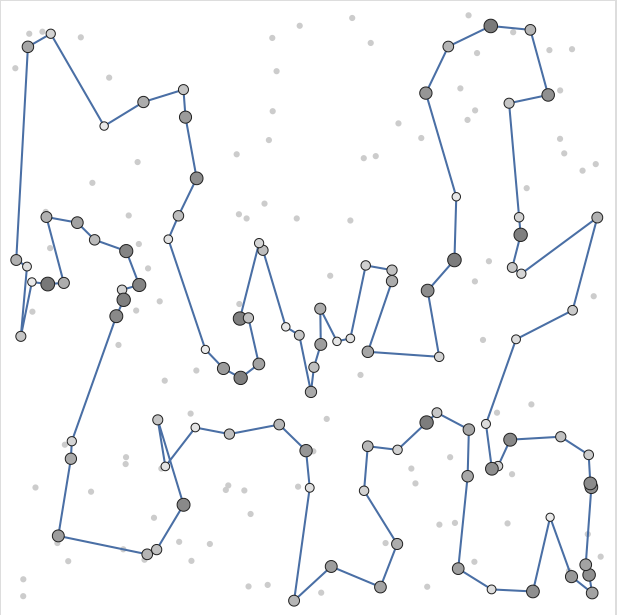
Lab 1: Random solution: objective 237131 (length 140636 + cost 96495)



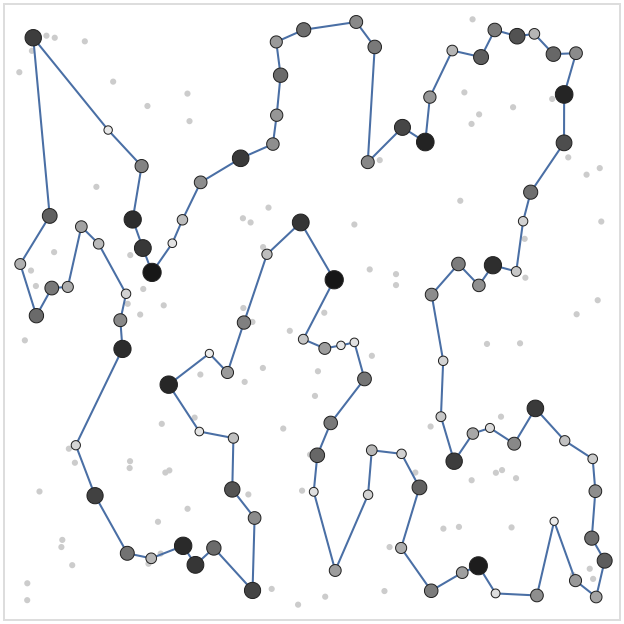
Lab 1: Nearest neighbor (end only): objective 84362 (length 31319 + cost 53043)



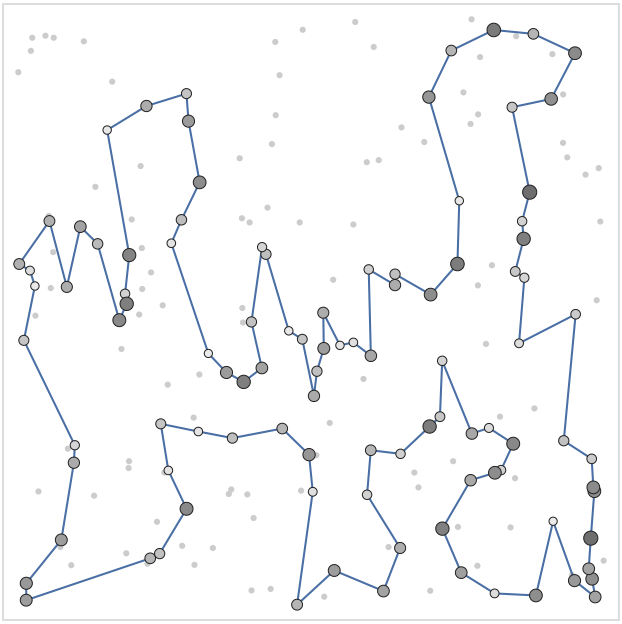
Lab 1: Nearest neighbor (any position): objective 72328 (length 21816 + cost 50512)



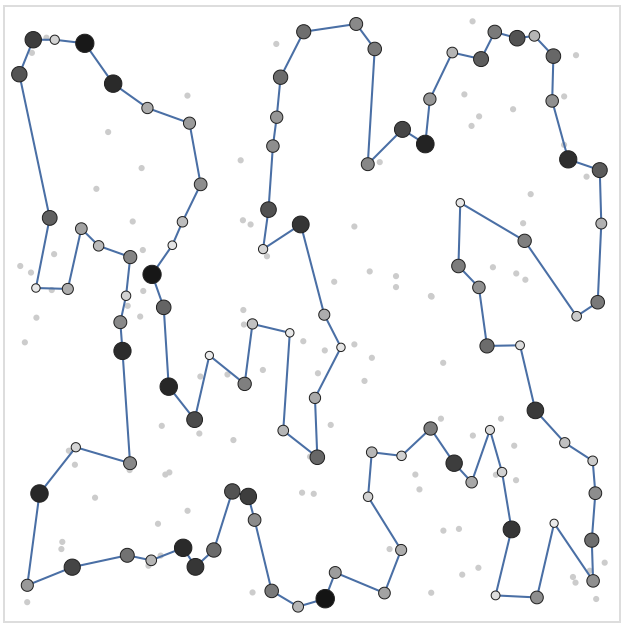
Lab 1: Greedy cycle: objective 72590 (length 21570 + cost 51020)



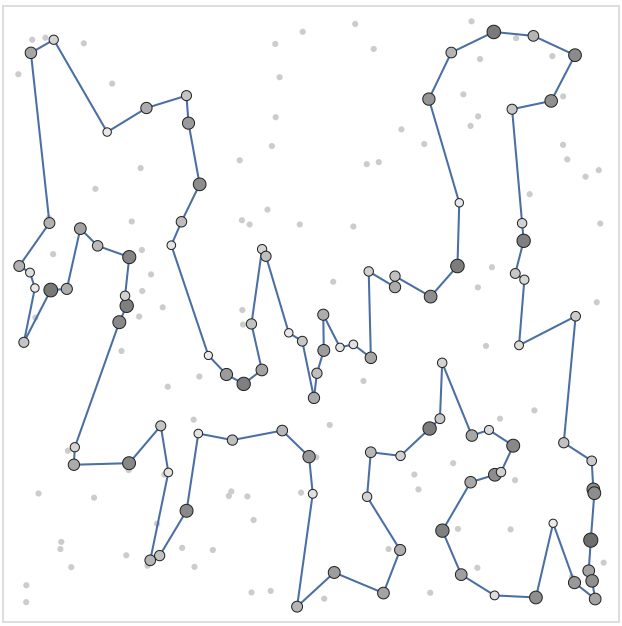
Lab 2: Nearest neighbor (any position), 2-regret: objective 111527 (length 19974 + cost 91553)



Lab 2: Nearest neighbor (any position), 2-regret + best change: objective 72419 (length 20127 + cost 52292)

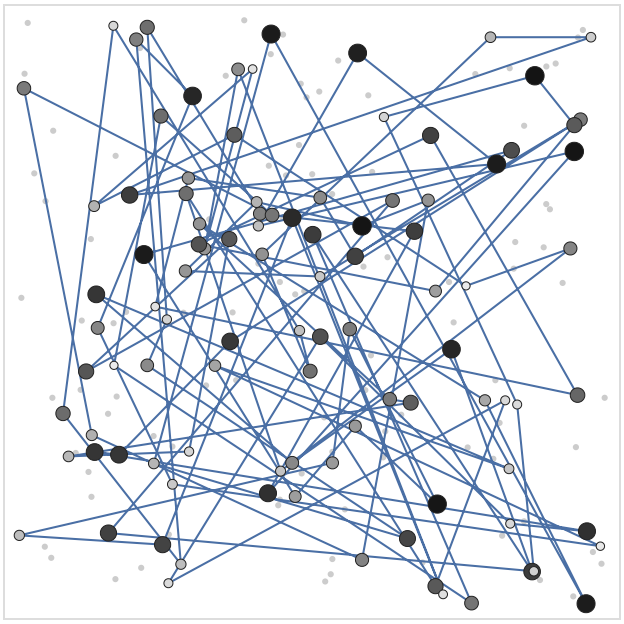


Lab 2: Greedy cycle, 2-regret: objective 112654 (length 21632 + cost 91022)

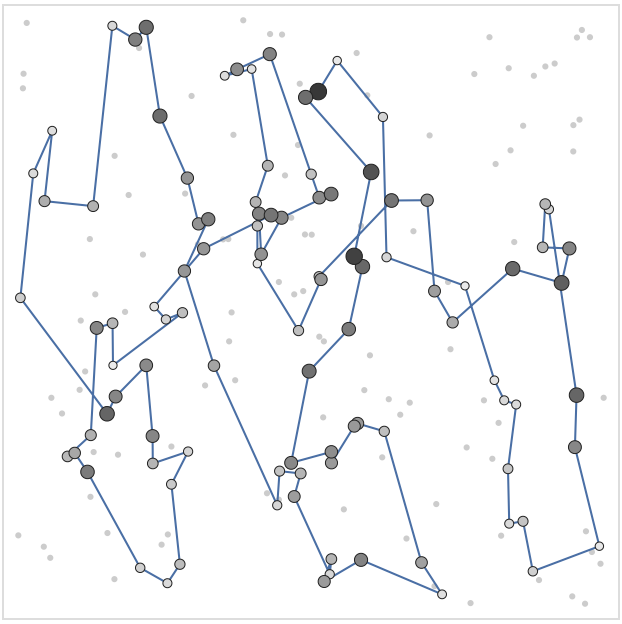


Lab 2: Greedy cycle, 2-regret + best change: objective 72011 (length 20248 + cost 51763)

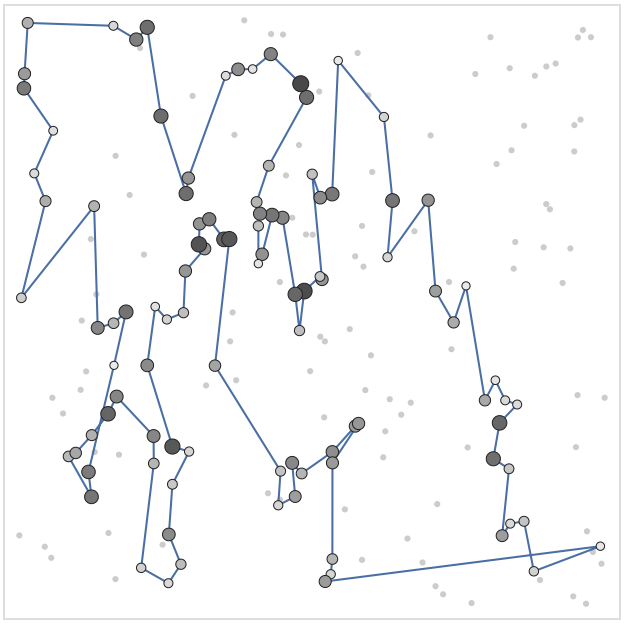
Instance TSPB



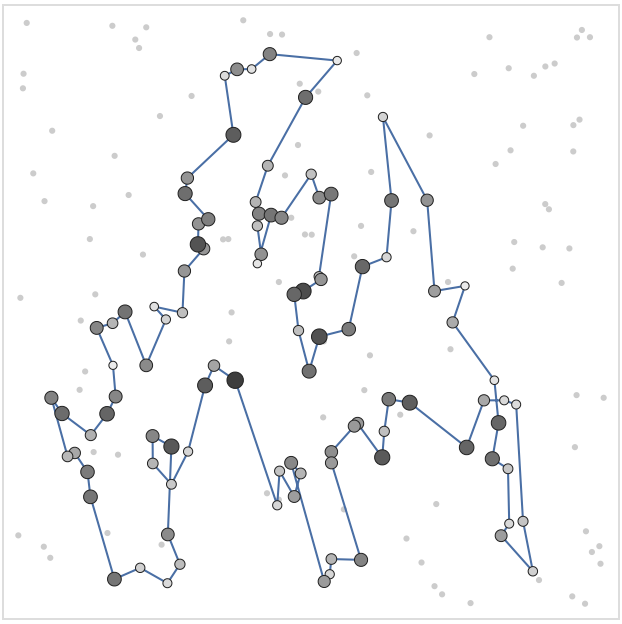
Lab 1: Random solution: objective 188214 (length 138112 + cost 50102)



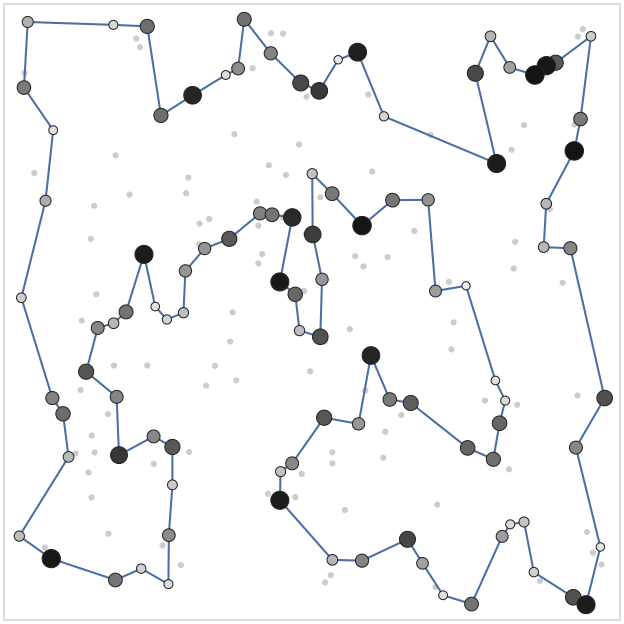
Lab 1: Nearest neighbor (end only): objective 56007 (length 24632 + cost 31375)



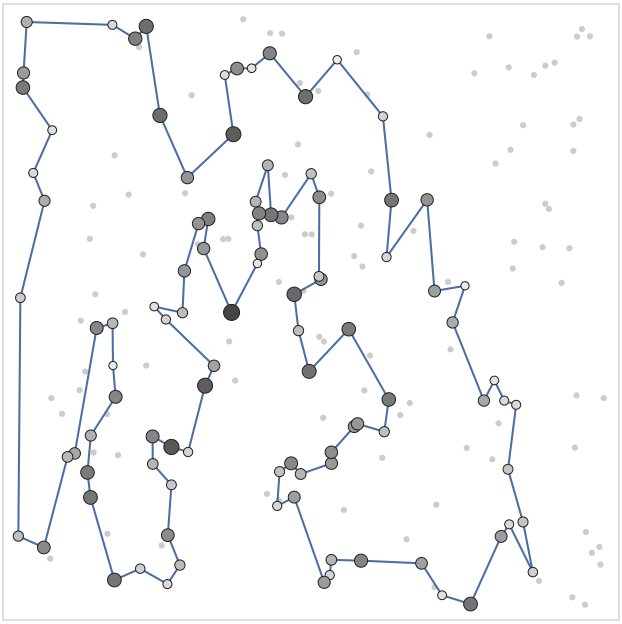
Lab 1: Nearest neighbor (any position): objective 52915 (length 20068 + cost 32847)



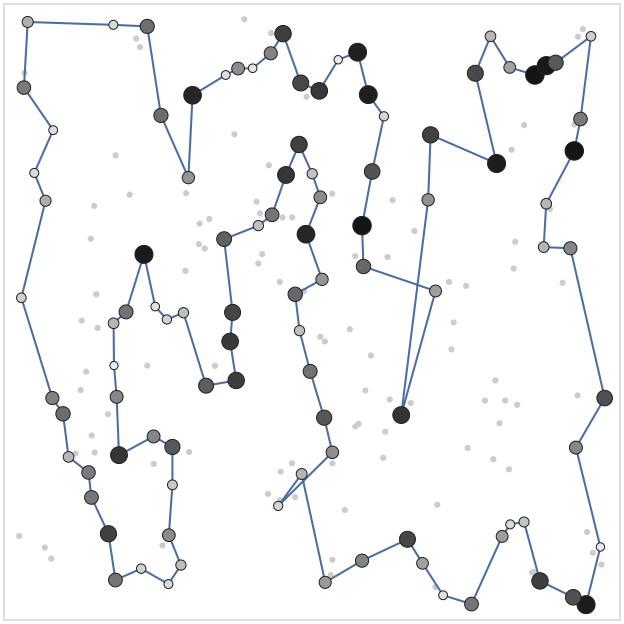
Lab 1: Greedy cycle: objective 53016 (length 16704 + cost 36312)



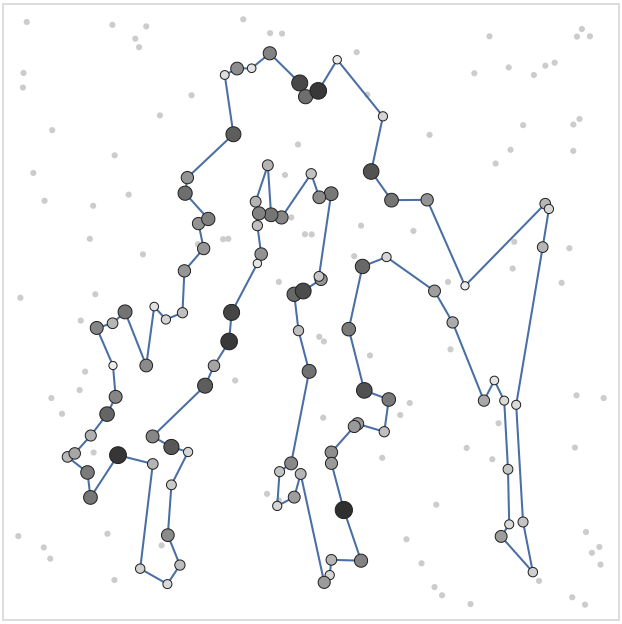
Lab 2: Nearest neighbor (any position), 2-regret: objective 69032 (length 21267 + cost 47765)



Lab 2: Nearest neighbor (any position), 2-regret + best change: objective 50485 (length 18835 + cost 31650)



Lab 2: Greedy cycle, 2-regret: objective 69455 (length 20736 + cost 48719)



Lab 2: Greedy cycle, 2-regret + best change: objective 52906 (length 16675 + cost 36231)

Best solutions (0-based node indices)

Instance TSPA

Lab 1: Random solution (objective 237131):
128, 34, 55, 103, 152, 57, 106, 85, 120, 112, 86, 43, 48, 96, 63, 38, 100, 61, 164, 195, 178, 194, 58, 117, 6, 158, 173, 40, 37, 121, 156, 192, 5, 101, 165, 1, 29, 131, 147, 60, 52, 170, 64, 54, 142, 93, 41, 88, 107, 159, 80, 68, 20, 36, 15, 22, 12, 146, 67, 116, 149, 189, 137, 133, 35, 99, 51, 30, 13, 154, 122, 75, 155, 33, 115, 162, 23, 87, 197, 56, 19, 78, 118, 82, 66, 182, 139, 190, 144, 161, 127, 65, 191, 50, 114, 45, 16, 26, 163, 129

Lab 1: Nearest neighbor (end only) (objective 84362):
54, 9, 33, 36, 177, 67, 144, 53, 135, 58, 134, 176, 174, 106, 89, 107, 57, 182, 151, 93, 50, 3, 1, 164, 178, 29, 136, 186, 184, 47, 159, 130, 153, 78, 88, 193, 170, 146, 75, 64, 150, 42, 113, 43, 85, 142, 8, 131, 84, 158, 192, 181, 114, 141, 180, 132, 76, 80, 48, 31, 63, 149, 168, 97, 127, 187, 30, 123, 152, 108, 60, 116, 49, 11, 119, 160, 83, 90, 51, 16, 35, 87, 45, 140, 52, 179, 185, 28, 102, 156, 157, 71, 91, 38, 100, 4, 112, 139, 115, 5

Lab 1: Nearest neighbor (any position) (objective 72328):
104, 129, 69, 127, 97, 168, 116, 149, 63, 11, 49, 160, 83, 123, 100, 152, 108, 60, 31, 48, 80, 91, 157, 71, 4, 76, 132, 180, 28, 102, 185, 35, 156, 51, 16, 67, 144, 140, 53, 179, 52, 177, 36, 33, 9, 54, 131, 174, 8, 164, 186, 136, 163, 29, 101, 178, 1, 169, 3, 50, 147, 47, 93, 106, 42, 89, 113, 130, 159, 146, 170, 153, 193, 78, 88, 107, 64, 57, 182, 151, 150, 176, 72, 134, 84, 58, 158, 192, 181, 114, 135, 73, 45, 87, 141, 187, 30, 189, 2, 171

Lab 1: Greedy cycle (objective 72590):
31, 116, 149, 168, 63, 11, 49, 60, 108, 152, 100, 90, 123, 83, 160, 97, 127, 69, 171, 2, 189, 30, 187, 141, 87, 45, 73, 135, 114, 181, 192, 158, 58, 84, 134, 72, 176, 150, 151, 182, 57, 64, 107, 88, 78, 193, 153, 170, 146, 159, 130, 113, 89, 42, 106, 93, 47, 147, 50, 3, 169, 1, 178, 101, 29, 163, 136, 186, 164, 142, 8, 174, 131, 54, 9, 33, 36, 177, 52, 179, 53, 140, 144, 67, 16, 51, 156, 35, 185, 102, 28, 180, 132, 76, 4, 71, 157, 91, 80, 48

Lab 2: Nearest neighbor (any position), 2-regret (objective 111527):
199, 82, 116, 119, 11, 49, 108, 152, 123, 160, 12, 97, 120, 92, 171, 166, 183, 10, 22, 98, 27, 87, 141, 23, 180, 28, 156, 16, 190, 74, 144, 179, 177, 36, 99, 41, 61, 135, 181, 58, 84, 134, 65, 158, 111, 182, 105, 57, 64, 107, 88, 78, 26, 75, 170, 159, 130, 117, 113, 106, 150, 0, 176, 174, 8, 142, 85, 128, 3, 1, 59, 46, 86, 55, 17, 29, 19, 163, 24, 136, 186, 109, 34, 62, 154, 39, 175, 115, 5, 112, 139, 18, 4, 76, 132, 68, 44, 14, 38, 80

Lab 2: Nearest neighbor (any position), 2-regret + best change (objective 72419):
171, 2, 189, 187, 30, 141, 87, 45, 73, 135, 114, 181, 192, 158, 58, 84, 134, 72, 176, 174, 150, 106, 113, 89, 42, 151, 138, 182, 57, 64, 107, 88, 78, 193, 153, 75, 170, 146, 159, 130, 47, 93, 50, 3, 169, 1, 59, 178, 101, 55, 29, 163, 136, 186, 164, 142, 8, 9, 54, 33, 131, 36, 177, 52, 179, 53, 140, 144, 67, 16, 51, 35, 185, 102, 28, 180, 132, 76, 4, 71, 157, 91, 80, 100, 123, 83, 160, 152, 108, 49, 60, 116, 149, 63, 168, 97, 127, 69, 129, 104

Lab 2: Greedy cycle, 2-regret (objective 112654):
190, 52, 177, 140, 61, 45, 67, 35, 102, 180, 145, 23, 162, 68, 132, 76, 4, 71, 91, 137, 126, 48, 199, 77, 82, 63, 49, 108, 152, 100, 123, 160, 12, 161, 97, 188, 129, 110, 92, 171, 166, 183, 10, 27, 70, 98, 25, 114, 118, 181, 192, 158, 58, 84, 134, 72, 0, 151, 106, 89, 124, 57, 64, 107, 193, 75, 170, 159, 130, 117, 93, 81, 85, 142, 164, 169, 47, 37, 147, 21, 155, 101, 17, 29, 19, 163, 24, 136, 186, 109, 34, 62, 154, 39, 175, 5, 112, 139, 133, 51

Lab 2: Greedy cycle, 2-regret + best change (objective 72011):
80, 91, 157, 71, 4, 76, 132, 180, 28, 102, 185, 35, 51, 16, 67, 144, 140, 53, 179, 52, 177, 36, 131, 33, 54, 9, 8, 142, 164, 186, 136, 163, 29, 55, 101, 178, 1, 169, 3, 50, 93, 47, 130, 159, 146, 170, 75, 153, 193, 78, 88, 107, 64, 57, 182, 138, 151, 42, 89, 113, 106, 150, 174, 176, 72, 134, 84, 58, 158, 192, 181, 114, 135, 73, 45, 87, 141, 189, 2, 171, 187, 30, 161, 127, 97, 160, 83, 123, 100, 152, 108, 49, 11, 168, 63, 149, 116, 60, 31, 48

Instance TSPB

Lab 1: Random solution (objective 188214):
80, 23, 16, 71, 163, 97, 140, 42, 136, 144, 107, 193, 172, 44, 12, 15, 105, 190, 160, 98, 151, 116, 103, 45, 147, 195, 177, 60, 110, 64, 153, 55, 73, 106, 188, 137, 17, 170, 50, 8, 13, 6, 88, 104, 0, 133, 38, 184, 61, 4, 113, 5, 68, 20, 111, 134, 129, 39, 191, 59, 148, 54, 63, 158, 96, 21, 198, 171, 95, 142, 1, 181, 27, 186, 34, 51, 161, 18, 75, 46, 130, 154, 56, 77, 176, 162, 139, 199, 57, 90, 2, 91, 185, 19, 168, 40, 132, 84, 66, 30

Lab 1: Nearest neighbor (end only) (objective 56007):
32, 89, 57, 76, 54, 127, 129, 90, 153, 20, 14, 2, 110, 118, 56, 114, 196, 10, 148, 143, 78, 192, 162, 104, 185, 184, 3, 151, 69, 172, 133, 120, 96, 121, 198, 48, 72, 46, 86, 117, 160, 136, 8, 21, 154, 140, 47, 170, 125, 177, 6, 132, 29, 197, 175, 134, 45, 60, 40, 71, 155, 97, 115, 180, 65, 28, 149, 145, 158, 116, 147, 42, 50, 130, 30, 183, 38, 64, 85, 59, 167, 51, 87, 179, 82, 88, 171, 99, 67, 24, 34, 181, 152, 73, 98, 27, 135, 75, 79, 156

Lab 1: Nearest neighbor (any position) (objective 52915):
110, 2, 14, 20, 74, 153, 182, 52, 90, 129, 127, 80, 54, 192, 162, 104, 76, 185, 57, 89, 86, 46, 72, 3, 184, 173, 151, 122, 117, 8, 160, 69, 172, 136, 133, 120, 156, 43, 48, 96, 198, 121, 130, 191, 50, 42, 147, 116, 37, 146, 13, 149, 28, 145, 65, 158, 177, 125, 26, 170, 197, 36, 132, 29, 6, 180, 115, 155, 71, 175, 134, 45, 157, 60, 40, 141, 97, 154, 140, 47, 38, 21, 176, 30, 183, 189, 168, 64, 59, 85, 51, 73, 167, 152, 34, 24, 181, 179, 87, 82

Lab 1: Greedy cycle (objective 53016):
49, 64, 33, 40, 60, 71, 155, 141, 157, 45, 134, 175, 62, 36, 197, 29, 132, 178, 103, 6, 180, 115, 170, 177, 125, 26, 97, 140, 154, 47, 38, 21, 176, 30, 183, 191, 130, 63, 121, 198, 96, 48, 89, 156, 120, 133, 136, 172, 160, 69, 8, 117, 72, 46, 86, 184, 3, 173, 122, 151, 98, 111, 27, 135, 76, 185, 57, 104, 162, 54, 192, 127, 52, 182, 153, 20, 74, 2, 14, 90, 129, 80, 150, 84, 66, 67, 31, 24, 34, 152, 181, 88, 179, 87, 82, 73, 167, 51, 59, 85

Lab 2: Nearest neighbor (any position), 2-regret (objective 69032):
57, 44, 89, 32, 43, 48, 58, 198, 121, 195, 50, 42, 116, 37, 13, 149, 145, 65, 178, 103, 132, 1, 108, 62, 175, 134, 157, 60, 141, 155, 113, 115, 19, 177, 125, 26, 161, 154, 140, 47, 38, 21, 168, 136, 8, 95, 128, 122, 151, 111, 3, 137, 72, 86, 0, 185, 104, 162, 54, 127, 129, 52, 182, 150, 84, 66, 35, 24, 25, 73, 59, 101, 179, 88, 163, 99, 171, 17, 74, 20, 14, 2, 93, 23, 110, 118, 131, 148, 10, 196, 193, 61, 105, 169, 165, 199, 70, 190, 194, 12

Lab 2: Nearest neighbor (any position), 2-regret + best change (objective 50485):
65, 145, 28, 149, 13, 146, 37, 116, 147, 42, 50, 130, 63, 121, 198, 96, 48, 156, 89, 57, 185, 76, 104, 162, 54, 192, 80, 127, 129, 90, 153, 14, 2, 20, 74, 17, 171, 99, 88, 179, 87, 82, 51, 85, 59, 73, 167, 181, 152, 34, 24, 67, 66, 27, 98, 151, 122, 3, 184, 46, 72, 117, 8, 120, 133, 136, 172, 160, 69, 124, 21, 183, 30, 38, 47, 154, 140, 64, 33, 40, 141, 155, 71, 60, 157, 45, 134, 175, 62, 36, 197, 6, 115, 170, 125, 177, 29, 132, 174, 1

Lab 2: Greedy cycle, 2-regret (objective 69455):
152, 25, 98, 151, 122, 3, 109, 46, 72, 166, 92, 8, 172, 189, 124, 4, 49, 33, 47, 140, 154, 161, 26, 125, 170, 115, 113, 155, 141, 60, 157, 45, 134, 175, 62, 91, 36, 197, 132, 103, 178, 65, 145, 28, 149, 13, 37, 116, 42, 50, 130, 195, 121, 198, 96, 48, 164, 43, 32, 89, 44, 83, 57, 79, 0, 135, 162, 119, 104, 77, 12, 194, 190, 70, 199, 165, 169, 105, 61, 193, 196, 10, 148, 131, 118, 110, 23, 93, 9, 14, 20, 74, 17, 171, 99, 163, 88, 82, 167, 85

Lab 2: Greedy cycle, 2-regret + best change (objective 52906):
60, 157, 45, 134, 175, 71, 113, 36, 197, 132, 29, 6, 180, 115, 170, 177, 125, 26, 97, 154, 140, 47, 38, 21, 30, 183, 191, 130, 63, 121, 198, 96, 48, 43, 156, 32, 89, 57, 79, 185, 104, 54, 196, 114, 10, 90, 14, 2, 74, 20, 153, 129, 127, 80, 192, 162, 76, 135, 27, 11, 66, 67, 24, 34, 152, 181, 126, 88, 179, 87, 82, 167, 51, 85, 59, 73, 98, 151, 122, 173, 3, 184, 86, 46, 72, 117, 8, 120, 133, 136, 172, 160, 69, 124, 4, 64, 33, 155, 141, 40

Source code

Source code of the compared labs:

- Lab 1: labs/lab1 (Python, C, C++, Rust)
- Lab 2: labs/lab2 (Python, C, C++, Rust)

Conclusions

TODO: write the conclusions.